

# CS3001-1 Computer Network and Communication

## Unit 1 — Combined MCQ Bank with Answers

Both question sets merged, duplicates removed, grouped by topic. Question on the left, answer with a one-line reason on the right.

### Network Basics, Topologies and Models

| Question                                                                                                                                                                                                   | Answer                                                                                                                                                       |
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| <b>1.</b> In a ____ communication, the media is dedicated.<br>a) point-to-point b) multipoint c) mesh d) None of the choices are correct                                                                   | <b>a) point-to-point</b><br>Point-to-point: one dedicated link between two devices; multipoint shares the link.                                              |
| <b>2.</b> In a fully connected mesh topology with six computers, we need ____ links.<br>a) 6 b) 36 c) 15 d) 30                                                                                             | <b>c) 15</b><br>$n(n-1)/2 = 6 \cdot 5/2 = 15$ duplex links (each node has $n-1 = 5$ ports).                                                                  |
| <b>3.</b> Which topology requires a multipoint connection?<br>a) Ring b) Bus c) Star d) Mesh                                                                                                               | <b>b) Bus</b><br>In a bus all devices tap into one shared backbone cable; ring, star and mesh use point-to-point links.                                      |
| <b>4.</b> What is the full form of OSI?<br>a) optical service implementation b) open service Internet c) open system interconnection d) operating system interface                                         | <b>c) open system interconnection</b><br>Open Systems Interconnection model (ISO), 7 layers.                                                                 |
| <b>5.</b> When a collection of various computers appears as a single coherent system to its clients, what is this called?<br>a) mail system b) networking system c) computer network d) distributed system | <b>d) distributed system</b><br>A distributed system hides the multiple machines behind one coherent interface (e.g. the Web).                               |
| <b>6.</b> Which one of the following is not a function of the network layer?<br>a) congestion control b) error control c) routing d) inter-networking                                                      | <b>b) error control</b><br>Error control belongs to the data-link and transport layers; the network layer does routing, logical addressing, internetworking. |
| <b>7.</b> Which layer is responsible for process-to-process delivery in a general network model?<br>a) session layer b) data link layer c) transport layer d) network layer                                | <b>c) transport layer</b><br>Transport layer (TCP/UDP) delivers between processes using port numbers; network layer is host-to-host.                         |
| <b>8.</b> What is the term for the data communication system within a building or campus?<br>a) MAN b) LAN c) PAN d) WAN                                                                                   | <b>b) LAN</b><br>LAN – a few km at most; PAN is personal (metres), MAN a city, WAN countries.                                                                |
| <b>9.</b> Which layer provides the services to the user?<br>a) physical layer b) presentation layer c) session layer d) application layer                                                                  | <b>d) application layer</b><br>Application layer (HTTP, FTP, SMTP, DNS) is the user interface to the network.                                                |
| <b>10.</b> Identify the layer which is responsible for data translating.<br>a) Network Layer b) Data Link Layer c) Presentation Layer d) Application Layer                                                 | <b>c) Presentation Layer</b><br>Presentation layer: translation (encoding), compression and encryption.                                                      |

| Question                                                                                                                                                                               | Answer                                                                                                                                  |
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| <b>11.</b> Which connection is necessary for a computer to join the internet?<br>a) internet society b) internet service provider c) different computer d) internet architecture board | <b>b) internet service provider</b><br>An ISP provides the physical/logical connection to the Internet backbone.                        |
| <b>12.</b> Currently ____ is responsible for the management of Internet domain names and addresses.<br>a) NIC b) ICANN c) ISOC d) IETF                                                 | <b>b) ICANN</b><br>ICANN – Internet Corporation for Assigned Names and Numbers (took over from InterNIC in 1998).                       |
| <b>13.</b> Which of the following is not one of the network criteria?<br>a) Performance b) Reliability c) Security d) Adaptability                                                     | <b>d) Adaptability</b><br>Forouzan's three criteria: performance, reliability, security.                                                |
| <b>14.</b> In ____ transmission, the channel capacity is shared by both communicating devices at all times.<br>a) simplex b) half-duplex c) full-duplex d) half-simplex                | <b>c) full-duplex</b><br>Full-duplex: both directions simultaneously (telephone); half-duplex: one direction at a time (walkie-talkie). |
| <b>15.</b> To deliver a message to the correct application program running on a host, the ____ address must be consulted.<br>a) port b) IP c) physical d) none of the above            | <b>a) port</b><br>Port (16-bit) identifies the process; IP identifies the host; MAC identifies the NIC on a link.                       |
| <b>16.</b> A single channel is shared by multiple signals by ____.<br>a) analog modulation b) digital modulation c) multiplexing d) phase modulation                                   | <b>c) multiplexing</b><br>Multiplexing (FDM, TDM, WDM) lets several signals share one link.                                             |

## Signals and Transmission Media

| Question                                                                                                                                                                                                                        | Answer                                                                                                              |
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| <b>17.</b> Frequency and period are ____.<br>a) inverse of each other b) proportional to each other c) the same d) are not related                                                                                              | <b>a) inverse of each other</b><br>$f = 1/T$ .                                                                      |
| <b>18.</b> As frequency increases, the period ____.<br>a) decreases b) increases c) remains the same d) None of the choices are correct                                                                                         | <b>a) decreases</b><br>$T = 1/f$ , so a higher frequency means a shorter period.                                    |
| <b>19.</b> ____ describes the position of the waveform relative to time 0.<br>a) Amplitude b) Time c) Frequency d) Phase                                                                                                        | <b>d) Phase</b><br>Phase (in degrees or radians) is the shift of the wave with respect to $t = 0$ .                 |
| <b>20.</b> The ____ of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.<br>a) frequency b) period c) bandwidth d) amplitude                                        | <b>c) bandwidth</b><br>$\text{Bandwidth} = f_{\text{max}} - f_{\text{min}}$ (in Hz).                                |
| <b>21.</b> ____ is a type of transmission impairment in which the signal loses strength due to the resistance of the transmission medium.<br>a) Attenuation b) Distortion c) Noise d) Decibel                                   | <b>a) Attenuation</b><br>Attenuation = loss of energy, measured in dB; overcome with amplifiers/repeaters.          |
| <b>22.</b> ____ is a type of transmission impairment in which the signal loses strength due to the different propagation speeds of each frequency that makes up the signal.<br>a) Attenuation b) Distortion c) Noise d) Decibel | <b>b) Distortion</b><br>Distortion: components arrive with different delays, so the composite signal changes shape. |

| Question                                                                                                                                                                 | Answer                                                                                                                                               |
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| <b>23.</b> Baseband transmission of a digital signal is possible only if we have a ____ channel.<br>a) low-pass b) bandpass c) low rate d) high rate                     | <b>a) low-pass</b><br>A digital signal has frequencies from 0 upward, so it needs a low-pass channel; bandpass channels need modulation (broadband). |
| <b>24.</b> Which transmission media provides the highest transmission speed in a network?<br>a) coaxial cable b) twisted pair cable c) optical fiber d) electrical cable | <b>c) optical fiber</b><br>Optical fibre: Gbps–Tbps, immune to EMI, lowest attenuation.                                                              |
| <b>25.</b> UTP is commonly used in ____.<br>a) DSL b) FTTP c) HTTP d) None of the mentioned                                                                              | <b>a) DSL</b><br>Unshielded twisted pair is the telephone-line medium used by DSL; FTTP uses fibre; HTTP is a protocol.                              |

## Encoding and Conversion

| Question                                                                                                                                                                                                           | Answer                                                                                                                                          |
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| <b>26.</b> ____ conversion involves three techniques: line coding, block coding, and scrambling.<br>a) Analog-to-digital b) Digital-to-analog c) Analog-to-analog d) Digital-to-digital                            | <b>d) Digital-to-digital</b><br>Digital data → digital signal: line coding (NRZ, Manchester...), block coding (4B/5B), scrambling (B8ZS, HDB3). |
| <b>27.</b> In ____, the change or lack of change in the level of the voltage determines the value of the bit.<br>a) NRZ-I b) NRZ-L c) NRZ-I or NRZ-L d) None of the choices are correct                            | <b>a) NRZ-I</b><br>NRZ-I (invert): a transition means 1, no transition means 0. NRZ-L uses the level itself.                                    |
| <b>28.</b> In Manchester and differential Manchester encoding, the transition at the middle of the bit is used for ____.<br>a) bit transfer b) baud transfer c) synchronization d) None of the choices are correct | <b>c) synchronization</b><br>The guaranteed mid-bit transition lets the receiver recover the clock (self-synchronizing).                        |
| <b>29.</b> The idea of RZ and the idea of NRZ-L are combined into the ____ scheme.<br>a) Manchester b) differential Manchester c) Manchester or differential Manchester d) None of the choices are correct         | <b>a) Manchester</b><br>Manchester = NRZ-L levels + RZ's mid-bit transition; differential Manchester combines RZ with NRZ-I.                    |
| <b>30.</b> The most common technique to change an analog signal to digital data is called ____.<br>a) PAL b) PCM c) sampling d) none of the above                                                                  | <b>b) PCM</b><br>Pulse Code Modulation: sampling → quantization → encoding (sampling alone is only the first step).                             |

## Data Link Layer, Error Detection and Ethernet

| Question                                                                                                                                                                                                                        | Answer                                                                                                                       |
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| <b>31.</b> A packet at the data-link layer is normally called a ____.<br>a) datagram b) message c) packet d) frame                                                                                                              | <b>d) frame</b><br>Frame (data-link), datagram/packet (network), segment (transport), message (application).                 |
| <b>32.</b> Which of the following best describes a single-bit error?<br>a) A single bit is inverted. b) A single bit per transmission is inverted. c) A single bit per data unit is inverted. d) All of the choices are correct | <b>c) A single bit per data unit is inverted.</b><br>Only one bit of a given data unit (byte/frame) changes from 0→1 or 1→0. |

| Question                                                                                                                                                                          | Answer                                                                                                                                                             |
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| <b>33.</b> If the ASCII character G is sent and the character D is received, what type of error is this?<br>a) Single bit b) Multiple bit c) Burst d) Recoverable                 | <b>c) Burst</b><br>G = 1000111, D = 1000100 → two adjacent bits (positions 0 and 1) changed → a burst error of length 2.                                           |
| <b>34.</b> Which error detection method involves polynomials?<br>a) CRC b) Simple parity check c) Two-dimensional parity check d) Checksum                                        | <b>a) CRC</b><br>CRC treats bit strings as polynomials (e.g. divisor 1011 = $x^3+x+1$ ) and uses modulo-2 division.                                                |
| <b>35.</b> In CRC, the quotient at the sender _____.<br>a) becomes the dividend at the receiver b) becomes the divisor at the receiver c) is the remainder d) is discarded        | <b>d) is discarded</b><br>Only the remainder (CRC bits) is appended to the dataword; the quotient is thrown away.                                                  |
| <b>36.</b> In CRC, if the dataword is 111111, the divisor 1010, and the remainder 110, what is the codeword at the receiver?<br>a) 111111011 b) 1010110 c) 111111110 d) 110111111 | <b>c) 111111110</b><br>Codeword = dataword followed by remainder: 111111 + 110 = 111111110 (assuming no error).                                                    |
| <b>37.</b> The ____ sublayer is responsible for the operation of the CSMA/CD access method and framing.<br>a) LLC b) MII c) MAC d) IP                                             | <b>c) MAC</b><br>MAC (Media Access Control) sublayer does framing, addressing and CSMA/CD; LLC handles flow/error control.                                         |
| <b>38.</b> What is the length of the MAC address?<br>a) 8 bits b) 16 bits c) 32 bits d) 48 bits                                                                                   | <b>d) 48 bits</b><br>48 bits = 6 bytes, written as 12 hex digits (e.g. 4A:30:10:21:10:1A).                                                                         |
| <b>39.</b> Which of the following can be an Ethernet multicast address?<br>a) 43:7B:6C:DE:10:00 b) 44:AA:C1:23:45:32 c) 46:56:21:1A:DE:F4 d) 48:32:21:21:4D:34                    | <b>a) 43:7B:6C:DE:10:00</b><br>Multicast ⇔ least-significant bit of the first byte is 1 ⇔ first byte is odd: 43 is odd (0100 0011); 44, 46, 48 are even (unicast). |
| <b>40.</b> Gigabit Ethernet has a data rate of ____ Mbps.<br>a) 10 b) 100 c) 1000 d) 10,000                                                                                       | <b>c) 1000</b><br>1 Gbps = 1000 Mbps (Standard Ethernet 10, Fast 100, 10-Gigabit 10,000).                                                                          |
| <b>41.</b> In the Ethernet, the ____ field is actually added at the physical layer and is not (formally) part of the frame.<br>a) CRC b) preamble c) address d) SFD               | <b>b) preamble</b><br>The 7-byte preamble (10101010...) is added by the physical layer for synchronization.                                                        |
| <b>42.</b> An ARP request is normally _____.<br>a) broadcast b) multicast c) unicast d) None of the choices are correct                                                           | <b>a) broadcast</b><br>The sender does not know the target MAC, so the request goes to FF:FF:FF:FF:FF:FF; the reply is unicast.                                    |